Related Work and Research Gap

NinaPro DB3 gesture recognition with graph neural networks · Task 1 (Track 1: GNN) · CSE475 Machine Learning, Fall 2026 · Group 02

Related Work

Table 1. Five GNN papers on sEMG gesture recognition

#	Paper	Dataset	Method (GNN type)	Metrics reported	Strengths	Limitations	Relation to our work
1	Lee, H., Jiang, M., Yang, J., Yang, Z., & Zhao, Q. (2025). *Decoding Gestures in Electromyography: Spatiotemporal Graph Neural Networks for Generalizable and Interpretable Classification.* IEEE TNSRE, 33, 404–419. DOI 10.1109/TNSRE.2024.3523943	Ninapro DB5 (10 intact subjects, 53 classes); CapgMyo A/B; UCI; BandMyo	GCN on a graph fixed to electrode layout (KNN-1, KNN-2, FC), with temporal links between windows. Fixed, layout-based graph.	Accuracy only: 99.4% intra-subject on DB5 (FC+SW); 92.4% inter-subject on DB5 (FC+SW).	Graph mirrors sensor layout, so edges can be explained. Evaluated across subjects and datasets. Reports reduced-sensor results.	Random 8:2 split within each subject. Intact subjects only. No amputee cohort. Inter-subject accuracy is well below intra-subject.	Closest method to ours (GCN on electrode layout). Its inter-subject DB5 figure is the cross-subject reference for our DB3 results.
2	Lee, H., Lee, S., Kim, J., Jung, H., Yoon, K. J., Gandla, S., Park, H., & Kim, S. (2023). *Stretchable array electromyography sensor with graph neural network for static and dynamic gestures recognition system.* npj Flexible Electronics, 7, 20. DOI 10.1038/s41528-023-00246-3	Own 8-channel stretchable array; 18 gestures (static and dynamic); 3 participants	Self-attention GNN: adjacency built per input window from cosine similarity, with temporal CNN/GLU blocks. Adaptive, learned graph.	Accuracy only: 97.76% at the best sensor position; 96.6% initially and 94.8% after 72 h.	Graph adapts to each input. Stable over 72 h and repeated re-attachment. Open access.	Three participants. Own hardware. Split within each participant (8:1:1). Not tested cross-subject or on a public benchmark.	Shows a learned adjacency as an alternative to a fixed electrode graph. Relevant to our graph-construction choice.

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3	Zhong, W., Zhang, Y., Fu, P., Xiong, W., & Zhang, M. (2023). *A Spatio-Temporal Graph Convolutional Network for Gesture Recognition from High-Density Electromyography (STGCN-GR).* M2VIP 2023. DOI 10.1109/M2VIP58386.2023.10413402; arXiv:2312.00553	HD-sEMG; 65 isometric hand gestures; 19 subjects (Malešević et al., 2021)	Spatio-temporal GCN: k-NN graph (k = 2) from Pearson correlation between 128 channels; temporal convolution with GLU. Correlation-learned graph.	Accuracy only: 91.07 ± 4.13% (5-fold CV, 250 ms windows).	Graph is learned from data. Large gesture set. Reports standard deviation across folds.	Folds are repetitions within the same subjects. HD-sEMG hardware. No cross-subject or amputee evaluation.	Correlation-based graph is a candidate construction for DB3's 12 channels. Compare against our fixed-layout graph.
4	Xu, X., & Jiang, H. (2023). *A Hybrid Model Based on ResNet and GCN for sEMG-Based Gesture Recognition.* Journal of Beijing Institute of Technology, 32(2), 219–229. DOI 10.15918/j.jbit1004-0579.2022.116	Ninapro DB1 (27 subjects, 52 movements)	ResNet feature extractor plus GCN over a fully connected muscle graph. GCN as one module in a hybrid CNN.	Accuracy only: 90.07% averaged over subjects.	Uses the Ninapro family, the same source as our DB3 data. Compares against 15 prior methods.	Fully connected graph ignores physical layout. GCN contribution not isolated from ResNet. Healthy subjects only.	Same Ninapro family as our data. Shows how a GCN can be combined with a CNN front end.
5	Vipulanandan, P., Premaratne, K., & Murthi, M. (2026). *A Graph Neural Network Model for Real-Time Gesture Recognition Based on sEMG Signals.* arXiv:2607.07850 (related DOI 10.1109/EMBC53108.2024.10781990)	Myo armband; 5 hand gestures; 8 healthy subjects	Two-layer fully connected GNN; graph per window from channel correlations, built only in detected-gesture windows. Correlation-based graph.	Accuracy only: 99% average; 48 ms per graph construction and prediction.	Real-time latency reported. Simple architecture.	Preprint. Five classes only. Split details not confirmed. Healthy subjects only.	Lightweight baseline for latency. Its 99% is on five classes and is not comparable to a 50-class DB3 result.

Research Gap

We did not find a GNN evaluated on the NinaPro DB3 dataset or on any transradial amputee cohort. The five papers in Table 1 use healthy-subject datasets (DB5, DB1, CapgMyo, Myo armband), an HD-sEMG array, or a small custom array. The searches covered web search results and the reference lists of the papers above; the absence of a DB3 GNN in these results is a finding about the sources reviewed, not a proof that no such work exists.

Two gaps follow from Table 1:

1. **Cross-subject generalisation.** Lee et al. (2025) report inter-subject accuracy of 92.4% on DB5, against 99.4% intra-subject. Most of the reviewed papers evaluate within subjects, so cross-subject performance of GNNs on large gesture sets is not well established. Our EDA supports this concern: subject identity dominates the UMAP embedding (silhouette +0.78), and gestures 1–8 overlap in PCA, t-SNE and UMAP.
2. **Amputee data.** None of the five papers uses an amputee cohort. Ninapro DB3 is an amputee benchmark, and whether GNNs transfer to amputee electrode placements has not been tested in these papers.

Our project addresses both gaps by testing a GNN on DB3 with subject-grouped 5-fold cross-validation, reporting macro-F1 and per-class recall alongside accuracy, and comparing against standard baselines.

References

Lee, H., Jiang, M., Yang, J., Yang, Z., & Zhao, Q. (2025). Decoding Gestures in Electromyography: Spatiotemporal Graph Neural Networks for Generalizable and Interpretable Classification. IEEE Transactions on Neural Systems and Rehabilitation Engineering, 33, 404–419. https://doi.org/10.1109/TNSRE.2024.3523943

Lee, H., Lee, S., Kim, J., Jung, H., Yoon, K. J., Gandla, S., Park, H., & Kim, S. (2023). Stretchable array electromyography sensor with graph neural network for static and dynamic gestures recognition system. npj Flexible Electronics, 7, 20. https://doi.org/10.1038/s41528-023-00246-3

Zhong, W., Zhang, Y., Fu, P., Xiong, W., & Zhang, M. (2023). A Spatio-Temporal Graph Convolutional Network for Gesture Recognition from High-Density Electromyography. In 2023 29th International Conference on Mechatronics and Machine Vision in Practice (M2VIP). https://doi.org/10.1109/M2VIP58386.2023.10413402

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Vipulanandan, P., Premaratne, K., & Murthi, M. (2026). A Graph Neural Network Model for Real-Time Gesture Recognition Based on sEMG Signals. arXiv:2607.07850. https://doi.org/10.48550/arXiv.2607.07850